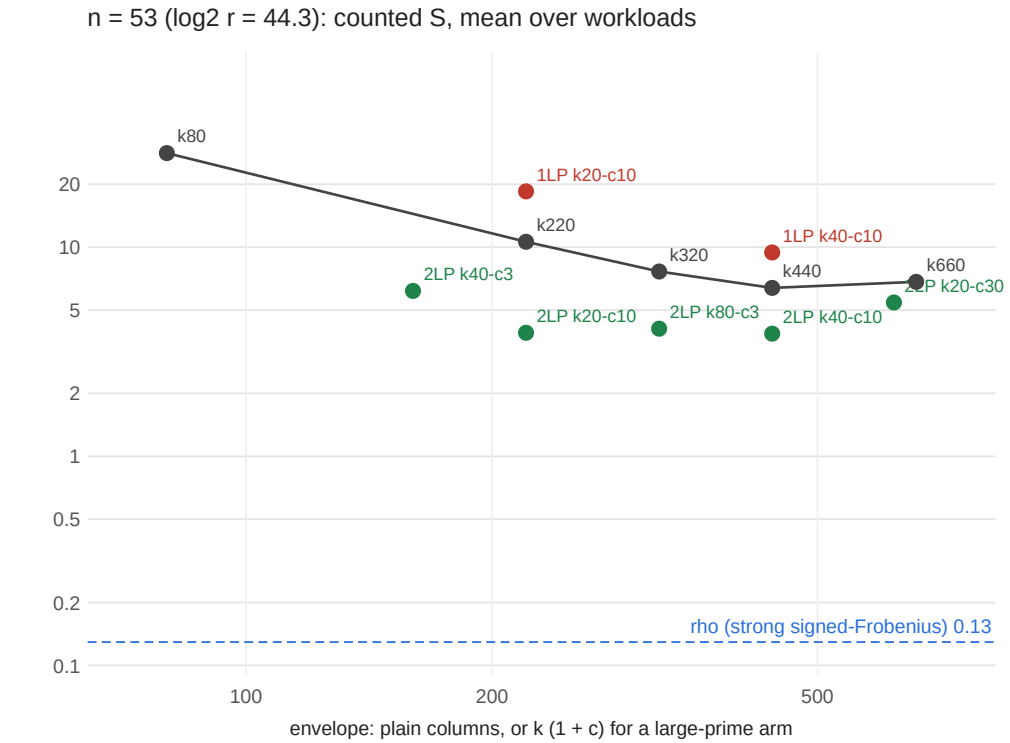
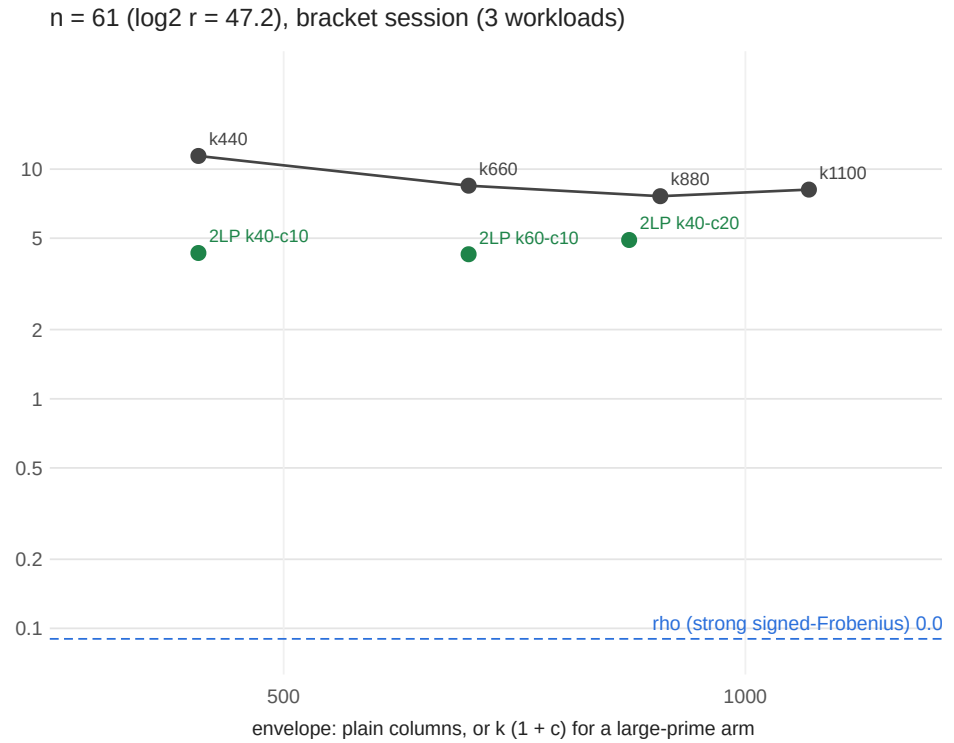
One- and two-large-prime collectors on the Koblitz compact-orbit pipeline, n = 53 and 61 (2026-10-29)

Atlas lead `LEAD-ICB-large-prime-variants`. Same ecbench harness, same planted-log workloads, counted S = GAE / sqrt(r); the arms differ only in the oracle's large-prime parameters and column counts. See `PROTOCOL.md`.





Findings

- **Verdict: a verified constant-factor improvement of the classic compact-orbit pair-table pipeline by the double large prime; the single large prime loses; rho is still 30-55 times ahead.** At n = 53 the best plain table (320, 440 and 660 columns bracket it at 7.65, 6.38 and 6.83) costs S = 6.38; the two-large-prime collector with a 40-column base and a domain of ten times that (`lp2-k40-c10`, the same 440-column envelope) costs 3.86, a paired ratio of 0.604 [0.599, 0.610] over 16 workloads. At n = 61 the best plain table is 880 columns (S = 7.62, bracketed by 660 and 1100 columns); `lp2-k60-c10` costs 4.26 (paired 0.559 [0.535, 0.578] over 3 workloads).
- **The order survives the pricing band for the best large-prime arms.** Native work (hash probes, eliminator row operations, the domain scan) is priced at 0, 0.1 and 1 GAE per operation for every arm, rho included. At n = 53 `lp2-k40-c10` against the best plain table: 0.604 / 0.635 / 0.730 (`ecbench band`: "lp2-k40-c10 < plain-k440" holds across the band). At n = 61: `lp2-k60-c10` 0.559 / 0.623 / 0.831 and `lp2-k40-c20` 0.644 / 0.651 / 0.675. Other two-large-prime arms win at weight 0 and lose at weight 1 (`lp2-k20-c10` at n = 53: 0.61 / 0.78 / 1.28; `lp2-k40-c10` at n = 61: 0.57 / 0.71 / 1.18): their native work is larger than their saved group operations.
- **The single large prime is worse than the plain table at equal envelope** (1.73 [1.64, 1.82] and 1.47 [1.42, 1.52] at n = 53), as at n = 41 in 2026-10-12.

- **Distance to the matched strong signed-Frobenius rho (S / S_{rho} at weights 0, 0.1, 1).** n = 53: best plain 49 / 49 / 48; best two-large-prime 30 / 31 / 35. n = 61: best plain (880 columns) 85 / 83 / 78; `lp2-k60-c10` 47 / 52 / 64, `lp2-k40-c20` 55 / 54 / 52. Rho is ahead of every arm at every weight; the gap is about halved, not closed.
- **It is a constant, not an exponent.** The best-arm ratio is 0.60 at n = 53 and 0.56 at n = 61, and the optimal large-prime base stays 40-60 columns while the plain optimum moves from 440 to 880. Two sizes cannot fit a slope; nothing here says the ratio grows.
- **Memory beside time** (median peak RSS of the harness's own measurement, wall time exploratory on a shared host): at n = 53 the best large-prime arm and the best plain table use the same memory (317 vs 313 MiB for `lp2-k40-c10` and `plain-k440`) and solve in a median 9.9 s against 13.3 s. At n = 61 `lp2-k60-c10` uses 604 MiB and 36.7 s against 1,149 MiB and 59.4 s for the best plain table (880 columns). Memory is not in S.
- **What this is not.** The leaderboard's Table A recipe for these curves is a different pipeline (aimed m = 3 collection with an m = 2 descent, in its own unit) at 7.03x (n = 53) and 11.5x (n = 61) of its reference, below the 30-55x here; the large-prime collector was not combined with that recipe, so no Table A row moves. The gain is inside the classic single-target pipeline, whose cost is dominated by the pair-table build.
- **Controls.** Every run verified (planted logarithm recovered, independent point check). The duplicate plain arm reproduces the plain arm's S exactly (A/A ratio 1.000 [1.000, 1.000]); the large-prime collector switched off (`large_primes=0`) reproduces it counter for counter. Replay certificate: `ecbench verify --replay-all` on every session.
- **Limits.** Counted S is deterministic, so the two rounds of a workload repeat each other; the independent units are workloads: 16 at n = 53 (two sessions of eight), 4 and 3 at n = 61 (two sessions). The n = 61 intervals are over three workloads and should be read as such. The 40-60 column optimum was found on the same curves as the confirmation (a grid selected by a pilot of two repeats on separate seeds). n = 61 first session (grid edge at 660 columns) is kept as evidence; the bracket session supplies the best plain table.

n = 53, icv1-f2m53-tm56619371-dac20a85

arm	verified	mean S	paired S vs equal-envelope control (95% interval)	priced ratio vs control, w = 0 / 0.1 / 1	S / rho, w = 0 / 0.1 / 1	median RSS MiB	median solve wall s
lp2-k40-c10	32/32	3.856	0.604 [0.599, 0.610] vs plain-k440	0.604 / 0.635 / 0.730	29.8 / 31.1 / 35.1	317	9.9
lp2-k20-c10	32/32	3.892	0.367 [0.358, 0.376] vs plain-k220	0.367 / 0.462 / 0.745	30.1 / 38.0 / 61.7	102	9.0
lp2-k80-c3	32/32	4.073	0.532 [0.525, 0.540] vs plain-k320	0.532 / 0.635 / 0.942	31.5 / 37.6 / 55.9	189	10.1
lp2-k20-c30	32/32	5.433	0.796 [0.790, 0.801] vs plain-k660	0.796 / 0.756 / 0.623	42.0 / 38.8 / 29.2	595	14.8
lp2-k40-c3	32/32	6.174	0.581 [0.563, 0.596] vs plain-k220	0.582 / 0.762 / 1.295	47.7 / 62.6 / 107.2	70	13.3
plain-k440	32/32	6.383	-	-	49.3 / 49.0 / 48.1	313	13.3
plain-k440-aa	32/32	6.383	1.000 [1.000, 1.000] vs plain-k440	1.000 / 1.000 / 1.000	49.3 / 49.0 / 48.1	313	13.3
plain-k660	32/32	6.828	-	-	52.8 / 51.3 / 46.9	596	16.9
plain-k320	32/32	7.651	-	-	59.1 / 59.2 / 59.3	173	14.6
lp1-k40-c10	32/32	9.435	1.474 [1.422, 1.525] vs plain-k440	1.478 / 1.955 / 3.422	72.9 / 95.8 / 164.6	317	24.3
plain-k220	32/32	10.605	-	-	82.0 / 82.2 / 82.8	100	18.3
lp1-k20-c10	32/32	18.490	1.728 [1.638, 1.823] vs plain-k220	1.744 / 2.315 / 4.017	142.9 / 190.2 / 332.4	102	42.2
plain-k80	32/32	28.179	-	-	217.8 / 218.0 / 218.7	35	43.7
rho-strong	32/32	0.129	-	-	-	30	0.1
lp0-k440	32/32	6.383	1.000 [1.000, 1.000] vs plain-k440	1.000 / 1.000 / 1.000	49.3 / 49.0 / 48.1	313	13.4

n = 61, session s1 (icv1-f2m61-t158598901-ab42b6c5, 4 workloads)

arm	verified	mean S	paired S vs equal-envelope control (95% interval)	priced ratio vs control, w = 0 / 0.1 / 1	S / rho, w = 0 / 0.1 / 1	median RSS MiB	median solve wall s
lp2-k40-c10	8/8	4.199	0.374 [0.364, 0.384] vs plain-k440	0.374 / 0.458 / 0.712	29.7 / 36.4 / 56.3	322	41.2
lp2-k80-c3	8/8	6.350	-	-	45.0 / 58.3 / 98.3	191	56.8
lp2-k20-c10	8/8	6.911	0.324 [0.312, 0.336] vs plain-k220	0.325 / 0.428 / 0.735	49.0 / 64.4 / 110.9	105	59.5
plain-k660	8/8	8.178	-	-	57.9 / 57.4 / 55.9	601	65.9

plain-k440	8/8	11.225	-	-	79.5 / 79.4 / 79.1	317	78.0
plain-k440-aa	8/8	11.225	1.000 [1.000, 1.000] vs plain-k440	1.000 / 1.000 / 1.000	79.5 / 79.4 / 79.1	317	82.8
plain-k220	8/8	21.271	-	-	150.7 / 150.7 / 151.0	102	135.2
rho-strong	8/8	0.141	-	-	-	30	0.2
lp0-k440	8/8	11.225	1.000 [1.000, 1.000] vs plain-k440	1.000 / 1.000 / 1.000	79.5 / 79.4 / 79.1	317	80.2

n = 61, session s2: bracket of the plain optimum (3 other workloads)

arm	verified	mean S	paired S vs equal-envelope control (95% interval)	priced ratio vs control, w = 0 / 0.1 / 1	S / rho, w = 0 / 0.1 / 1	median RSS MiB	median solve wall s
lp2-k60-c10	6/6	4.262	0.503 [0.488, 0.513] vs plain-k660	0.503 / 0.553 / 0.707	47.4 / 51.6 / 64.4	603	36.7
lp2-k40-c10	6/6	4.311	0.378 [0.371, 0.389] vs plain-k440	0.378 / 0.464 / 0.724	47.9 / 58.8 / 91.3	315	36.1
lp2-k40-c20	6/6	4.909	0.644 [0.628, 0.654] vs plain-k880	0.644 / 0.651 / 0.675	54.5 / 54.0 / 52.3	1149	43.9
plain-k880	6/6	7.620	-	-	84.7 / 82.9 / 77.5	1149	59.4
plain-k1100	6/6	8.145	-	-	90.5 / 87.3 / 77.6	2254	72.8
plain-k660	6/6	8.475	-	-	94.2 / 93.4 / 91.1	594	59.2
plain-k440	6/6	11.400	-	-	126.7 / 126.5 / 126.2	310	72.4
plain-k440-aa	6/6	11.400	1.000 [1.000, 1.000] vs plain-k440	1.000 / 1.000 / 1.000	126.7 / 126.5 / 126.2	310	72.0
rho-strong	6/6	0.090	-	-	-	24	0.1
lp0-k440	6/6	11.400	1.000 [1.000, 1.000] vs plain-k440	1.000 / 1.000 / 1.000	126.7 / 126.5 / 126.2	310	71.1